Cross-Border E-Commerce Collaboration Platform

INFO 5100: Application Engineering & Development

Group 07 | Northeastern University | Prof. Bugrara

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Problem Statement

Cross-border e-commerce is a multi-party problem.

Suppliers, logistics providers, and e-commerce platforms each operate independently — yet the fulfillment of a single international order requires cooperative effort across all three enterprises.

Fragmented Operations

Suppliers, logistics, and platforms share no common system. Orders are passed via email or manual processes, causing delays and errors.

Cross-Border Complexity

International orders span different countries, currencies, and time zones, making coordination even harder without a unified system.

No Risk Visibility

High-value orders and delayed shipments go undetected until it's too late — resulting in customer complaints and financial loss.

Our solution: A unified Java Swing platform enabling real-time, role-based collaboration across enterprises.

Ecosystem Structure — Why Each Enterprise is Essential

Global E-Commerce Network

Supplier Enterprise China

- Product Management Org
- Order Processing Org

Roles: Supplier Manager · Supplier Staff

Why essential:
Sources all products and fulfills procurement requests from the platform.

Logistics Enterprise Global

- Shipment Organization
- Delivery Organization

Roles: Logistics Coordinator · Delivery Staff

Why essential:
Handles international shipping, customs coordination, and last-mile delivery.

Platform Enterprise USA

- Order Management Org
- Customer Service Org

Roles: Platform Order Mgr · Customer Service Agent

Why essential:
Connects customers to the supply chain and manages the entire order lifecycle.

Stakeholders & Roles (8 Unique Operational Roles)

#	Role	Responsibility	Work Area	Enterprise
1	**Supplier Manager**	Manage products (CRUD + restock); process work requests	*SupplierWorkAreaJPanel*	Supplier
2	**Supplier Staff**	Process fulfillment assignments, update inventory	*SupplierWorkAreaJPanel*	Supplier
3	**Logistics Coordinator**	Manage shipments (CRUD); track transit; assign delivery staff	*LogisticsWorkAreaJPanel*	Logistics
4	**Delivery Staff**	View assigned shipments; mark delivered or failed	*LogisticsWorkAreaJPanel*	Logistics
5	**Platform Manager**	Manage orders (create, ship, deliver, cancel); returns; reports	*PlatformWorkAreaJPanel*	Platform
6	**Customer Service Agent**	View all orders; resolve customer tickets	*PlatformWorkAreaJPanel*	Platform
7	**Data Analyst**	View order summary, country summary, and KPI dashboards	*DataAnalystJPanel*	Platform
8	**Customer**	Place orders; request returns	*CustomerJPanel*	Platform

Core Use Cases & Work Requests (6 Work Requests)

1

ProcurementRequest

Platform → Supplier Enterprise

Cross-enterprise ✓

2

RestockRequest

Platform → Supplier Enterprise

Cross-enterprise ✓

3

ShipmentRequest

Supplier → Logistics Enterprise

Cross-enterprise ✓

4

ReturnRequest

Platform → Supplier Enterprise

Cross-enterprise ✓

5

FulfillmentAssignment

Product Mgmt Org → Order Processing Org

Cross-organization ✓

6

DeliveryAssignment

Shipment Org → Delivery Org

Cross-organization ✓

Role-Based Work Areas — Different Roles, Different Dashboards

UI Layer

LoginJFrame · WorkAreaJPanels
OrderJPane
l
·
ReportViewerJPanel · SystemOverviewJPanel

▼

Business

Network · Enterprise · Organization
Order · Product · WorkRequest

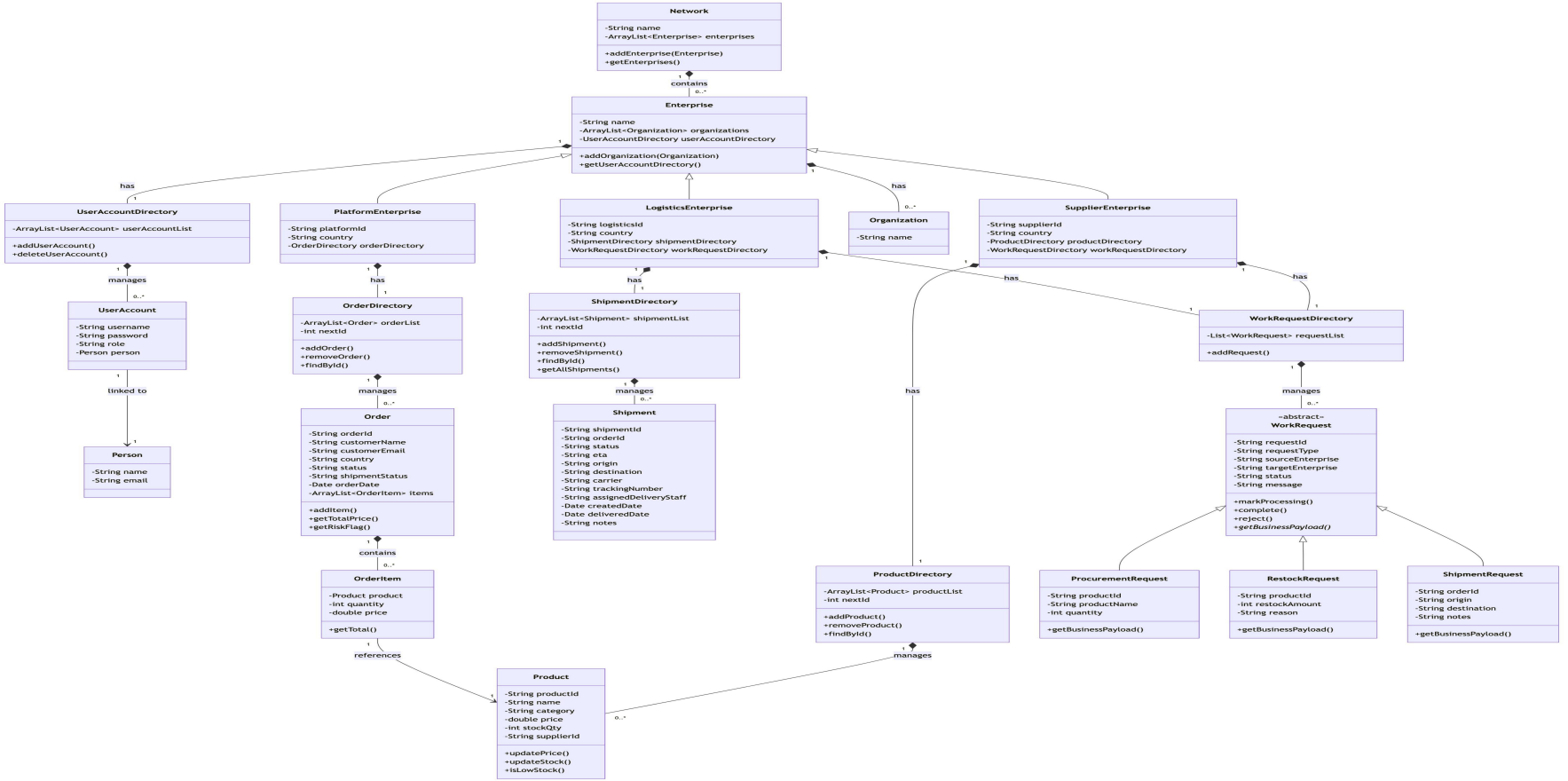
▼

Data

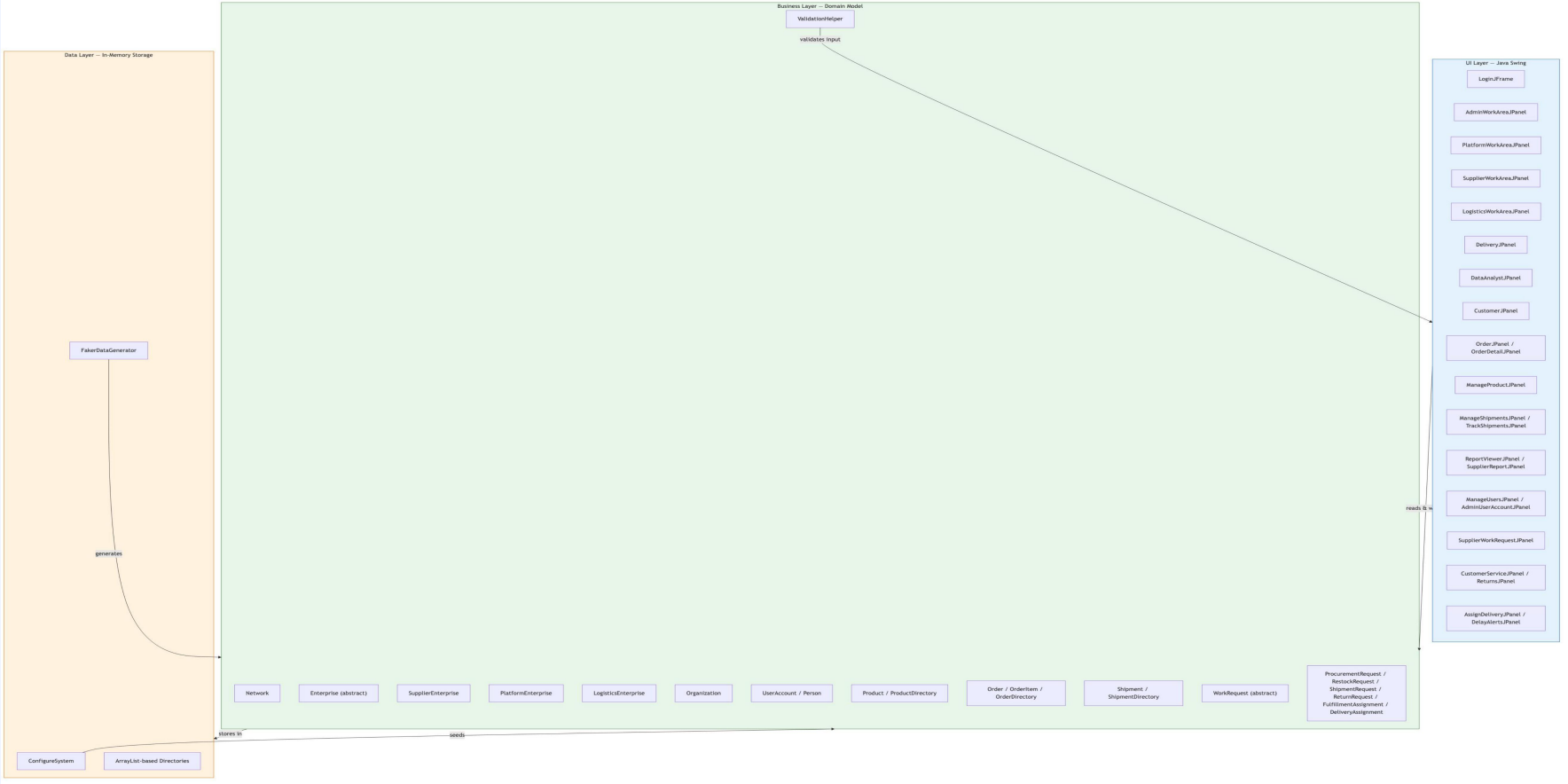
ConfigureSystem · FakerDataGenerator
ArrayList in-memory directories

Role-based auth · CRUD · Faker · Form validation

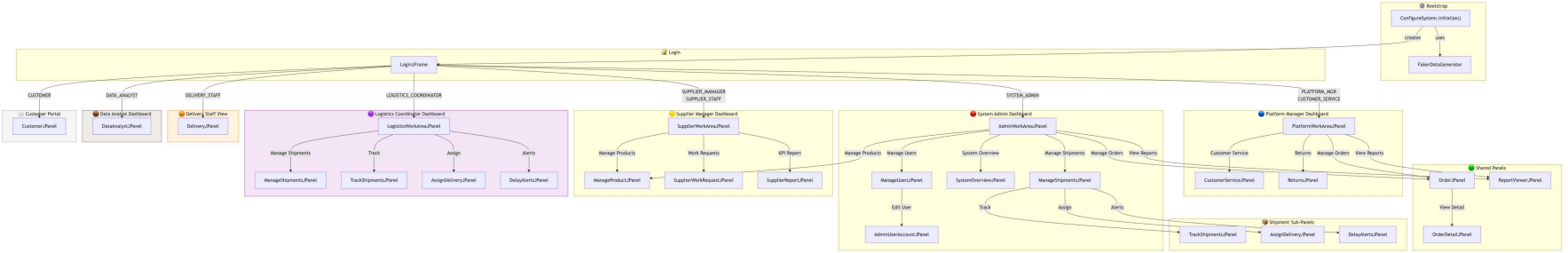
UML Class Diagram — Attributes, Methods & Relationships



High-Level Component Diagram — 3-Layer Architecture



UI Navigation Flow — Role-Based Routing & Panel Architecture



Order Management — Status Lifecycle & Risk Detection

Status Lifecycle

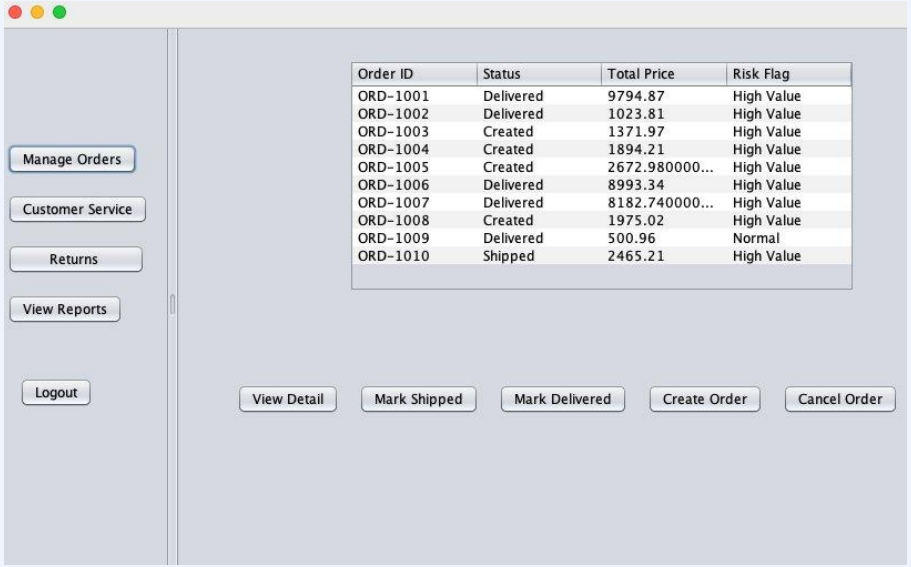
Created → Shipped → Delivered → Returned
Work requests: Created → Processing → Completed / Rejected

Risk Flag Column

High Value (>$1,000) and Delay Risk (>3 days unshipped)
auto-detected in every order row

Full CRUD

Create Order · Mark Shipped · Mark Delivered · Cancel Order
All with null checks and validation



Reporting Module & Advanced Feature

Summarized Reporting View

Tab 1: Order Summary

Order ID · Status · Total Price · Shipment Status · Risk Flag
Displayed per order — sortable JTable

Tab 2: Shipment Summary

Counts by status: Pending / Shipped / Delivered / Others
Total revenue + High Risk order count

Tab 3: Country Summary

Per-country breakdown: Order Count · Revenue · High Risk Orders
Grouped using HashMap for analytics

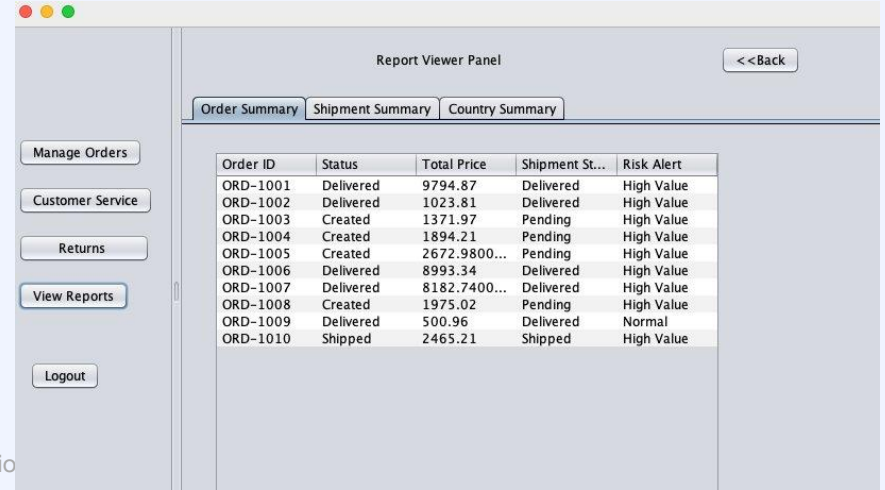
Implemented as JTabbedPane with 3 tabs — real-time data from PlatformEnterprise.OrderDirectory

Advanced Feature: Smart Risk Detection

High Value · price > $1,000

Delay Risk · unshipped > 3 days

Normal · all criteria passed



Order ID	Status	Total Price	Shipment St...	Risk Alert
ORD-1001	Delivered	9794.87	Delivered	High Value
ORD-1002	Delivered	1023.81	Delivered	High Value
ORD-1003	Created	1371.97	Pending	High Value
ORD-1004	Created	1894.21	Pending	High Value
ORD-1005	Created	2672.9800...	Pending	High Value
ORD-1006	Delivered	8993.34	Delivered	High Value
ORD-1007	Delivered	8182.7400...	Delivered	High Value
ORD-1008	Created	1975.02	Pending	High Value
ORD-1009	Delivered	500.96	Delivered	Normal
ORD-1010	Shipped	2465.21	Shipped	High Value

Individual Contributions — Team of Three

Jianting Dong

Platform + Orders + Reports + Admin

- LoginJFrame — auth & role-based routing
- AdminWorkAreaJPanel — network-wide admin
- PlatformWorkAreaJPanel — full work area
- OrderJPanel — CRUD + status transitions
- CustomerServiceJPanel + ReturnsJPanel
- ReportViewerJPanel — 3-tab reporting
- SystemOverviewJPanel — network KPIs
- DataAnalystJPanel + CustomerJPanel
- PlatformEnterprise + OrderDirectory
- ConfigureSystem + FakerDataGenerator + ValidationHelper

Yiru Chen

Supplier + Products + Work Requests

- SupplierWorkAreaJPanel — work area
- ManageProductJPanel — full CRUD
- SupplierReportJPanel — KPI dashboard
- SupplierWorkRequestJPanel — request mgmt
- SupplierEnterprise + ProductDirectory
- Product.java — full model & validation
- WorkRequest (abstract base class)
- ProcurementRequest + RestockRequest
- FulfillmentAssignment + ReturnRequest

Jiachuan Li

Logistics + Shipments + Delivery

- LogisticsWorkAreaJPanel — work area
- ManageShipmentsJPanel — CRUD + sub-panels
- TrackShipmentsJPanel — shipment tracking
- AssignDeliveryJPanel — staff assignment
- DelayAlertsJPanel — risk alerts
- DeliveryJPanel — staff delivery view
- Shipment.java + ShipmentDirectory
- LogisticsEnterprise class
- ShipmentRequest + DeliveryAssignment

Shared by all: Network · Enterprise (base) · Organization · UserAccount · Person · ConfigureSystem scaffold

Live Demonstration

20-Minute Demo Plan · All team members demonstrate their individual contributions

1

System Admin

Login → System Overview → Manage Users (Add/Delete/Edit) → Manage Products → Manage Shipments

2

Platform Manager

Login → Create Order → View Orders (Risk Flag) → Mark Shipped → Mark Delivered

3

Customer Service

Login → View Customer Issues → Resolve Ticket → View updated status

4

Supplier Manager

Login → Manage Products (CRUD + Restock) → View Work Requests → Mark Processing → Complete

5

Logistics Coordinator

Login → Manage Shipments → Track → Assign Delivery Staff → View Delay Alerts

6

Delivery Staff

Login → View Assigned Shipments → Mark Delivered / Mark Failed → Logout

7

Data Analyst

Login → Order Summary → Country Summary → KPI Dashboard

8

Report Dashboard

View Reports → Order Summary · Shipment Summary · Country Summary · Risk Detection flags